

Enhanced Gas Absorption in the Ionic Liquid 1-*n*-Hexyl-3-methylimidazolium Bis(trifluoromethylsulfonyl)amide ([hmim][Tf₂N]) Confined in Silica Slit Pores: A Molecular Simulation Study

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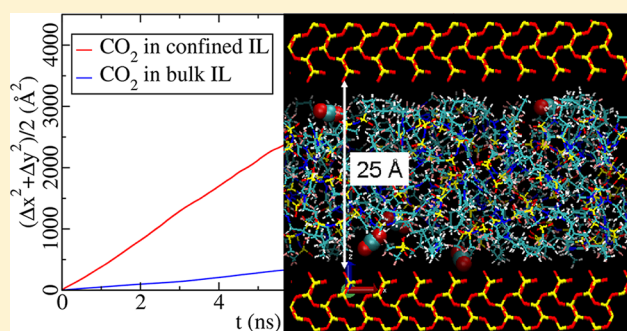
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S Supporting Information

ABSTRACT: Two-dimensional $NP_{xy}T$ and isostress-osmotic ($N_2P_{xy}Tf_1$) Monte Carlo simulations were used to compute the density and gas absorption properties of the ionic liquid (IL) 1-*n*-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide ([hmim][Tf₂N]) confined in silica slit pores (25–45 Å). Self-diffusivity values for both gas and IL were calculated from NVE molecular dynamics simulations using both smooth and atomistic potential models for silica. The simulations showed that the molar volume of [hmim][Tf₂N] confined in 25–45-Å silica slit pores is 12–31% larger than that of the bulk IL at 313–573 K and 1 bar. The amounts of CO₂, H₂, and N₂ absorbed in the confined IL are 1.1–3 times larger than those in the bulk IL because of the larger molar volume of the confined IL compared to the bulk IL. The CO₂, N₂, and H₂ molecules are generally absorbed close to the silica wall where the IL density is very low. This arrangement causes the self-diffusivities of these gases in the confined IL to be 2–8 times larger than those in the bulk IL at 298–573 K. The solubilities of water in the confined and bulk ILs are similar, which is likely due to strong water interactions with [hmim][Tf₂N] through hydrogen bonding, so that the molar volume of the confined IL plays a less important role in determining the H₂O solubility. Water molecules are largely absorbed in the IL-rich region rather than close to the silica wall. The self-diffusivities of water correlate with those of the confined IL. The confined IL exhibits self-diffusivities larger than those of the bulk IL at lower temperatures, but smaller than those of the bulk IL at higher temperatures. The findings from our simulations are consistent with available experimental data for similar confined IL systems.



1. INTRODUCTION

Ionic liquids (ILs) have received intense attention because of their unique properties, such as low vapor pressure, non-flammability, wide liquid temperature range, wide electrochemical window, high thermal stability, and high solvating capacity for both polar and nonpolar compounds.^{1,2} Generally, ILs are highly viscous, with viscosity values typically 2–3 orders of magnitude larger than that of water. Although highly viscous ILs are favored in certain applications such as stationary phases for gas–liquid chromatography,³ high viscosity causes many serious problems in chemical processing, such as requiring much more power to pump the IL and significantly slowing mass and heat transfer for gas absorption and reactions in ILs. Consequently, new devices and/or novel methods have to be developed to improve mass and heat transfer in ILs before ILs are adopted in large-scale industrial applications.

To improve mass transfer for gas absorption in ILs, molecular simulations were conducted that showed that the CO₂ self-diffusion coefficient increases by a factor of 17 in 1-*n*-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide

([hmim][Tf₂N]) confined in a (20,20) carbon nanotube (CNT) (diameter = 27 Å) compared to that in the unconfined bulk IL. Additionally, the confined IL exhibits self-diffusivity 1–2 orders of magnitude larger than that of the bulk IL.⁴ The enhanced self-diffusivity values for both gas and confined IL were mainly ascribed to the smoothness of the CNT and larger molar volume of the confined IL compared to the bulk IL.⁴ Recently, Hung and co-workers completed a series of theoretical studies of ILs confined in nanomaterials. They found that the IL self-diffusivity under confinement is very heterogeneous because of different packing of the IL in nanomaterials.⁵ The confined IL was found to diffuse faster than the bulk IL depending on the IL loading and the pore size of the nanomaterial.⁶ Intrigued by this theoretical work, Iacob et al. experimentally determined that the self-diffusivity coefficient of confined [bmim][BF₄] in silica nanopores (pore

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size = 75–104 Å) is increased by 2 orders of magnitude compared to that of the bulk IL.⁷ They attributed the enhanced IL self-diffusivity to the changes in ion packing and the reduction of the density of the IL in small pores,⁷ which is also consistent with the results of our simulations.⁴ Recently, Close et al.⁸ performed an experimental investigation of the CO₂ absorption behavior in confined ILs. They found that the CO₂ permeability in ILs supported in alumina nanopores (200 and 1000 Å) could be increased by a factor of 4 compared to the permeability in the bulk IL.

Despite this progress, some important questions are yet to be addressed. For example, even though the values of CO₂ permeability (solubility × diffusivity) in alumina-pore-confined ILs are larger than those in the corresponding bulk ILs,⁸ it is unclear how confinement affects the CO₂ solubility and diffusivity. Although the confined IL exhibits much larger self-diffusivity values than the bulk IL at low temperatures, Iacob et al. also found that, at higher temperatures, the silica-confined IL exhibits smaller self-diffusivities than the bulk IL.⁷ We did not observe these temperature-dependent confinement effects on IL self-diffusivity in our previous simulations of CNT-confined ILs.⁴ The one-dimensional CNT interacts more strongly with the ILs than the two-dimensional graphite and silica systems. It would be interesting to investigate whether ILs confined in weakly interacting silica pores still exhibit enhanced gas solubility and diffusivity behaviors similar to those of ILs confined in the strongly interacting CNT. Answering these questions will help researchers to enhance IL properties in a variety of applications including reaction media, nonvolatile electrolytes, separation solvents, and gas separation. Additionally, the investigation of the effects of confinement in a simple silica slit pore on gas absorption would help pave the way to modeling of the IL confined in more realistic and more complicated MCM-41 pores.

In this work, we addressed such questions using molecular modeling. We systematically studied CO₂, H₂, N₂, and H₂O gas absorption in [hmim][Tf₂N] confined in silica slit pores with widths between 25 and 45 Å. Pore sizes of 25–45 Å were chosen because [hmim][Tf₂N] exhibits several cation and anion layers in these pores, which allows the effects of pore size on the IL properties for gas absorption to be studied. Additionally, simulation results in these silica pores were compared with previous work on the same IL confined in a (20,20) CNT with a diameter of 27 Å.⁴ Finally, these silica pore sizes are similar to the pore size values for MCM-41 (20–50 Å), which has been modeled atomistically.⁹ Both gas solubility and self-diffusivity in the confined IL were compared against the values in the bulk IL. Additionally, the self-diffusivity values of the confined and bulk IL were also compared with each other at low and high temperatures. CO₂, H₂, N₂, and H₂O were studied because these gases represent the major gas components from which CO₂ is captured. Simulation results were also compared with available experimental data for similar confined IL systems.

2. THEORY AND SIMULATION METHODS

2.1. $NP_{xy}T$ Ensemble. The $NP_{xy}T$ ensemble (constant number of particles, constant xy -component pressure tensor, and constant temperature) was used to compute the density of the pure [hmim][Tf₂N] confined within two-dimensional silica slit pores. In this ensemble, the fundamental thermodynamic equation describing a system of N pairs of cations and anions adsorbed inside a slit pore is given by

$$dG = -S dT + \mu dN + T_{zz} dH - A d(T_{xy}H) \quad (1)$$

where $G = \mu N$ is the Gibbs free energy; μ is the chemical potential of the adsorbate (the adsorbed IL); T is the temperature; S is the entropy; H is the slit pore width; A is the area of the wall; T_{zz} and T_{xy} are the stresses in the directions perpendicular and parallel to the walls, respectively, which are equal to the negative values of the corresponding pressure tensor components. By specifying values for the variables T , N , H , and $T_{xy} = -P_{xy}$, we performed $NP_{xy}T$ Monte Carlo (MC) simulations to compute the average molar volume of the confined IL. The acceptance rules for the $NP_{xy}T$ ensemble are very similar to those used in generic NPT MC simulations, except for the change of area A (see the Supporting Information for details), rather than volume as in the NPT ensemble.

2.2. Isostress-Osmotic ($N_2P_{xy}Tf_1$) Ensemble. We used the isostress-osmotic ensemble to calculate the amounts of gas absorbed in the confined IL. The isostress-osmotic ensemble is a combination of the grand isostress¹⁰ and osmotic¹¹ ensembles. The fundamental thermodynamic equation for the isostress-osmotic ensemble is given by

$$dG_2 = -S dT - N_1 d\mu_1 + T_{zz} dH - A d(T_{xy}H) + \mu_2 dN_2 \quad (2)$$

where $G_2 = \mu_2 N_2$. Subscripts 1 and 2 denote the gas solute and the IL solvent molecules, respectively. The meanings of the other symbols in eq 2 are the same as in eq 1. By specifying T , μ_1 (f_1), T_{xy} ($-P_{xy}$), N_2 , and H , where f_1 is the fugacity of the gas solute, we performed isostress-osmotic MC simulations. Note that the pressure tensor component P_{xy} was set to be equal to the bulk gas pressure. Implementations for the isostress-osmotic ensemble are similar to those for the osmotic ensemble, in which T , f_1 , P , and N_2 are specified.¹² The only difference is that the pressure P in the osmotic ensemble is replaced by the pressure tensor components P_{xy} in the isostress-osmotic ensemble. Additionally, in the isostress-osmotic ensemble, it is also necessary to specify the slit pore width H .

The continuous fractional component (CFC) method^{12,13} was used to insert and delete gas solute molecules in the isostress-osmotic ensemble. The acceptance rules in the CFC isostress-osmotic MC ensemble are very similar to those for the CFC osmotic ensemble.¹² Specifically, in the isostress-osmotic MC simulations, there are three types of moves, namely, the thermal move, the area A move, and the λ move, where λ is the coupling strength between the fractional gas solute molecule and the other solute and solvent molecules. Details about this method can be found in our previous works.^{12,14,15} In simulations, the interaction between the fractional gas solute molecules and the silica wall was not scaled. Instead, when an attempt was made to insert a fractional gas solute molecule inside the slit pore at a position separated by less than 0.5 Å from the wall, the attempted insertion was simply rejected, and the old configuration was collected for sampling and property calculations.

2.3. Simulation Details. **2.3.1. Classical Force Field.** A classical force field was used to simulate the IL, gas solute, IL–gas, and adsorbate (gas, IL)–adsorbent interactions. The interaction energy for the system is given by

$$\begin{aligned} \mathcal{V}(\mathbf{r}) = & \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_\theta(\theta - \theta_0)^2 \\ & + \sum_{\text{dihedrals}} k_\chi[1 + \cos(n_\chi\chi - \delta_0)] + \sum_{\text{impropers}} k_\psi(\psi - \psi_0)^2 \\ & + \sum_{i=1}^{N-1} \sum_{j=i+1}^N \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{r_{ij}} \right\} + U_{\text{sf}} \end{aligned} \quad (3)$$

where the symbols have their conventional meanings,¹⁶ and U_{sf} describes the interaction between the adsorbate and the two opposite silica walls.

U_{sf} was computed using two models, namely, smooth and atomistic potential models. The smooth potential model was used in our in-house MC and molecular dynamics (MD) software. In MD

simulations using the smooth potential, a simple reflection from the wall (reflective boundary conditions) rather than randomizing of the velocities of particles after repulsion from the wall (diffusive boundary conditions)¹⁷ was used. The adsorbate atom interactions in a slit pore with two walls were calculated as $U_{sf} = \phi_{sf}(z) + \phi_{sf}(H - z)$, and $\phi_{sf}(z)$ is given by the 9–3 potential¹⁸

$$\phi_{sf}(z) = \frac{2\pi\rho_s\sigma_{sf}^3\epsilon_{sf}}{3} \left[\frac{2}{15} \left(\frac{\sigma_{sf}}{z} \right)^9 - \left(\frac{\sigma_{sf}}{z} \right)^3 \right] \quad (4)$$

where z is the distance between the atom and the wall and ρ_s is the silica number density. Cross-parameters σ_{sf} and ϵ_{sf} were calculated from the standard Lorentz–Berthelot mixing rules. The parameters for silica are $\sigma_s = 3.0$ Å, $\epsilon_s = 0.8$ kJ/mol, and $4\pi\rho_s = 0.5/\text{Å}^3$.¹⁹

In MD simulations using the NAMD program,²⁰ the atomistic silica potential model was used to obtain more realistic estimates for the self-diffusivity values. U_{sf} was calculated by explicitly summing all of the Lennard-Jones (LJ) and electrostatic (ELEC) interactions between the adsorbates and the Si and O atoms of the silica wall. The Si and O atom positions were fixed during simulations by using the *fixedAtoms* utility provided in the NAMD program. The charges and LJ parameters for Si and O atoms in the silica structure were developed in this work (see section 3). Two atomistic silica slit pores (25 and 45 Å) were constructed. Each silica pore consisted of two slabs and contained 980 Si atoms and 1960 O atoms on the two opposite walls. The silica wall had dimensions of $45.99 \text{ Å} \times 45.99 \text{ Å}$ in the xy plane parallel to the silica walls. The thickness of each silica slab was about 7.7 Å. The procedure to construct silica slit pores is described in detail in the Supporting Information. In this work, we did not investigate the surface chemistry such as the surface hydroxylation of the silica.

In CFC MC simulations, the LJ parameters of the fractional molecule were scaled using a functional form proposed by van Gunsteren et al.²¹ to aid insertion when computing the interaction between the fractional molecule and other adsorbate molecules. A switching function was turned on at 10.5 Å and turned off at 12.0 Å for the LJ potential. A Verlet neighbor list with a 13.5-Å radius was used. The intramolecular ELEC and LJ interactions for atoms separated by exactly three consecutive bonds were scaled by 0.5 and were neglected for atoms separated by less than three consecutive bonds. The classical force-field parameters for H_2O , CO_2 , N_2 , H_2 , $[\text{TF}_2\text{N}]^-$ anion, and $[\text{hmim}]^+$ cation were obtained from previous works.^{15,22,23} To evaluate the accuracy of the force field for $[\text{hmim}][\text{TF}_2\text{N}]$, the simulated densities of the IL were compared with the experimental values. At 313 and 373 K, the simulated densities differed by only 0.1–0.3% from the experimental values.²⁴ At 423 K and 400 bar, the simulated density agreed with the experimental value²⁵ within the simulation uncertainty. At 573 K, no experimental density is available for comparison. The experimental densities in the temperature range of 313–373 K²⁴ were linearly extrapolated to 573 K. The simulated density at 573 K differed by 2.3% from the experimental extrapolation value. The force-field parameters for the IL satisfactorily reproduced the experimental densities at 313–573 K. The Fennel and Gezelter shift force (FGSF)²⁶ method was used to accelerate the calculation of the ELEC interaction in MC simulations. A cutoff value of 12.0 Å and a damping parameter of $\kappa = 0.2022 \text{ Å}^{-1}$ were used in the FGSF method. The FGSF method was also validated against the standard Ewald method in this work (Supporting Information). In MD simulations, the standard Ewald method was used in the locally developed MD code, and the particle mesh Ewald (PME) method was used in the NAMD program to compute the ELEC interactions. In this work, a pseudo-two-dimensional system was applied to calculate ELEC interactions in the Ewald-type and FGSF methods by using a large simulation box size in the z direction compared to the silica slit pore width.

2.3.2. Monte Carlo Simulations. The $NP_{xy}T$ simulations were performed with a pressure tensor P_{xy} of 1 bar and in the temperature range of 313–573 K for 60 pairs of $[\text{hmim}]^+$ cations and $[\text{TF}_2\text{N}]^-$ anions confined in silica slit pores (25–45 Å). The simulation box size in the z direction was set to be 50–60 Å to minimize the interactions between adsorbate molecules in the primary cell and image cells. Simulations included several million steps of equilibration followed by

about 20 million steps of production runs. During the equilibration, various MC moves were tuned to achieve roughly 50% acceptance rates.⁴ These moves were thermal equilibration by hybrid Monte Carlo^{4,15,27} (HMC) (90%) and area A changes (10%). The values in parentheses represent the fixed probabilities during the run.

In the $N_2P_{xy}Tf_1$ simulations, the number of IL molecules was set to 60 pairs. For CO_2 , the temperatures and pressure tensors were set to be in the ranges of 333–573 K and 4–83.6 bar, respectively. For H_2 , the temperatures and pressure tensors were 313–573 K and 50–300 bar, respectively. For N_2 , the temperatures and pressure tensors were 298–333 K and 40–500 bar, respectively. In the case of H_2O , the temperature was set to 373 K, and the pressure tensors were 0.05–0.2 bar. Equilibration runs of typically 2 million moves were carried out during which various MC moves were tuned to achieve roughly 50% acceptance rates. These moves were thermal equilibration by HMC (40%), λ moves (50%), and area A moves (10%). Production runs of about 20 million steps were typically used. The CFC bias factor¹² was adjusted during the equilibration stage to achieve, as closely as possible, a uniform probability distribution of λ . Optimization of these bias factors was done using the Wang–Landau²⁸ updating scheme. The gas fugacity f_1 was calculated from the Peng–Robinson equation of state¹⁵ for CO_2 , N_2 , and H_2O and from MD simulations for H_2 .²³ Note that P_{xy} was set to be equal to the pressure in the gas phase.

The CFC grand canonical¹² MC simulations were performed to compute CO_2 , H_2 , N_2 , and H_2O adsorption into empty 25–45-Å silica slit pores (without confined ILs) at temperatures and pressures similar to those used in the $N_2P_{zz}Tf_1$ simulations. Simulation details in this case were very similar to those for $N_2P_{zz}Tf_1$, except that there is no A move in the grand canonical ensemble. The slab wall lengths in the x and y directions were set to be 40–200 Å.

2.3.3. Molecular Dynamics Simulations. Molecular dynamics simulations in the NVT and NVE ensembles were performed using both the in-house software for the smooth silica potential and the NAMD program²⁰ for the atomistic silica potential. The time step was 0.5–1.0 fs. Simulations were performed for 60–155 $[\text{hmim}][\text{TF}_2\text{N}]$ IL molecules confined in 25–45-Å silica slit pores. The densities of the confined IL in the MD simulations were obtained from $NP_{xy}T$ MC simulations at $P_{xy} = 1$ bar. Simulations were typically carried out for 5 ns of equilibration followed by 10-ns production runs. The self-diffusivity in the xy directions was calculated from the NVE MD trajectories using a procedure similar to the one described previously^{23,29}

$$D_{\text{self},xy} = \lim_{t \rightarrow \infty} \frac{1}{4t} \langle |x(t) - x(0)|^2 + |y(t) - y(0)|^2 \rangle \quad (5)$$

Self-diffusivities were calculated using both the smooth and atomistic silica potential models. Most of the results reported in the next section were obtained using the smooth silica potential model, unless the atomistic silica potential model was explicitly indicated for use in simulations.

Note that, in a binary mixture consisting of species A and B , the self-diffusion coefficients of A and B are defined as the diffusion coefficients of the individual label molecules A^* and B^* , that is, D_{A^*} and D_{B^*} , when there is no concentration gradient of $(A + A^*)$ or $(B + B^*)$.³⁰ In experiments, D_{A^*} and D_{B^*} can be measured using isotopes. In simulations, the self-diffusion coefficients D_{A^*} and D_{B^*} are calculated from the mean squared displacements of A and B ,³¹ which is also consistent with the definition of self-diffusivity in mixtures.

3. RESULTS AND DISCUSSION

3.1. Solid–Fluid Interaction. The interactions between the C atom of CO_2 and the silica slit pore, graphite slit pore, and CNT were obtained using the 9–3 potential for graphite³² and silica pores and the atomistic potential for the CNT.⁴ The results are shown in Figure S2 of the Supporting Information. In the case of slit pores, two well depths were positioned close to the walls. The well depths were the same for the 25-, 35-, and 45-Å silica slit pores, and they were about 1.8 and 4 times

smaller than for the graphite slit pore and the (20,20) CNT, respectively. All other atoms such as the O atom in CO_2 ; the F, O, S, C, and N atoms in the $[\text{Tf}_2\text{N}]^-$ anion; and the H, N, and C atoms in the $[\text{hmim}]^+$ cation exhibited the same behavior. The adsorbate atom interactions with the 25-Å silica slit pore were about 2 and 4 times weaker than those with the 25-Å graphite slit pore and the (20,20) CNT, respectively.

3.2. Thermodynamic Properties for Pure $[\text{hmim}][\text{Tf}_2\text{N}]$ Confined in Silica Slit Pores. 3.2.1. *Molar Volume and Density of $[\text{hmim}][\text{Tf}_2\text{N}]$ Confined in Silica Slit Pores.* Because molar volume is important in determining gas solubility in ILs,²³ the molar volume and density of confined $[\text{hmim}][\text{Tf}_2\text{N}]$ were calculated, and the results are reported in Table 1 and

Table 1. Molar Volume (V_m) of $[\text{hmim}][\text{Tf}_2\text{N}]$ Confined in Silica Pores (25–45 Å) Obtained from $NP_{xy}T$ Monte Carlo Simulations at 313–573 K and a Pressure Tensor (P_{xy}) of 1 bar, along with Computed Averages of the Simulation Box Lengths (L_x and L_y) in the xy Plane^a

pore width (Å)	T (K)	V_m (cm^3/mol)	$L_x = L_y$ (Å)
25	313	414.9(5)	40.66(3)
25	423	444(1)	42.07(6)
25	573	534(5)	46.1(2)
35	313	375.0(9)	32.67(4)
35	423	410(1)	34.17(4)
35	573	497(2)	37.61(8)
45	313	369(1)	28.61(4)
45	423	397.9(7)	29.68(2)
45	573	473(1)	32.36(4)

^aUncertainty in the last digit given in parentheses.

Figure S3 (Supporting Information). Note that different sets of simulations starting from different initial configurations were run with the same temperatures and pressure tensors; the molar volumes among these different simulations varied by only 1.7%. These results imply that the simulations were long enough to obtain good estimates of the molar volume and density of the confined IL.

The densities of $[\text{hmim}][\text{Tf}_2\text{N}]$ confined in silica slit pores were found to be smaller than those of the bulk IL. In silica slit pores with pore widths between 25 and 45 Å, the confined IL exhibited a 12–31% smaller density than the bulk IL at 313–573 K and 1 bar. The density of the confined IL increased when the slit pore width was increased. For example, when the pore width was increased from 25 to 35 Å, the density of the confined IL increased by 7–11%. When the pore width was further increased from 35 to 45 Å, the confined IL became more densely packed. As the silica pore continues to become larger and larger, the density of the confined IL is expected to approach the value for the bulk IL. Interestingly, the IL in the (20,20) CNT exhibits a density smaller than that in the 25-Å-wide silica slit pore.

3.2.2. Structure of $[\text{hmim}][\text{Tf}_2\text{N}]$ Confined in Silica Slit Pores. The local density of the center of mass of the IL confined in silica slit pores was also calculated, and the local density profiles are shown in Figure 1 and Figure S4 (Supporting Information). In the 25-Å-wide silica slit pore, the $[\text{hmim}]^+$ cation exhibits two layers at about $z = \pm 5$ Å. The distance between the two neighboring $[\text{hmim}]^+$ layers is about 10 Å. The $[\text{Tf}_2\text{N}]^-$ anion exhibits three layers. Two layers are positioned at $z = \pm 7$ Å, and the third layer occurs at the center. The distance between the two neighboring $[\text{Tf}_2\text{N}]^-$ layers is

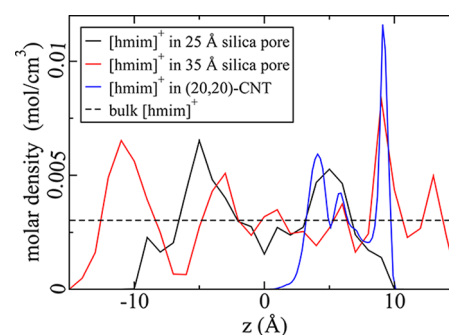


Figure 1. Local molar density of the center of mass of $[\text{hmim}]^+$ cations in silica slit pores. The results were obtained from $NP_{xy}T$ Monte Carlo simulations at $T = 313$ K and $P_{xy} = 1$ bar. For comparison, the cation densities in bulk $[\text{hmim}][\text{Tf}_2\text{N}]$ ²³ and $[\text{hmim}][\text{Tf}_2\text{N}]$ confined in a (20,20) carbon nanotube (CNT)⁴ at 313 K and 1 bar are also shown. Legend: black and red solid lines, 25- and 35-Å silica slit pores, respectively; blue solid line, (20,20) CNT; and black dashed line, bulk ionic liquid. Note that longer simulations would lead to smoother and more symmetric density distributions. In the case of the CNT, the z axis represents the radial distance of the C atom from the CNT tube center. This radial distance is greater than zero, and only one-half of the profile is shown for the CNT.

about 7 Å. The confined IL exhibits a smaller local density than the bulk IL in certain regions. For example, in the three regions of $-12.5 \text{ Å} \leq z \leq -6.5 \text{ Å}$, $-2.0 \text{ Å} \leq z \leq 3.0 \text{ Å}$, and $6.8 \text{ Å} \leq z \leq 12.5 \text{ Å}$, the confined $[\text{hmim}]^+$ exhibits a smaller local density than the bulk. It is due to the smaller local IL densities in these regions that lead to a smaller overall density for the confined IL compared to the bulk IL (Figure S3, Supporting Information).

In the case of 35-Å-wide silica slit pore, there are four cation layers; the distance between the neighboring $[\text{hmim}]^+$ layers was estimated to be 7.5 Å (Figure 1). Five layers of $[\text{Tf}_2\text{N}]^-$ occur and the distance between the neighboring anion layers is about 6 Å (Figure S4, Supporting Information). The distances between the neighboring cation/anion layers in the 35-Å-wide slit pore are smaller than in the 25-Å-wide slit pore. This arrangement leads to a more densely packed IL structure and larger densities of the confined IL in the 35-Å-wide slit pore compared to the 25-Å-wide slit pore. Note that the peak heights in the 25- and 35-Å-wide slit pores are similar. This similarity is partly due to the same solid–fluid potential well depth for the 25- and 35-Å-wide silica slit pores (Figure S2, Supporting Information).

The density peak heights for the cation/anion layers in the silica slit pores are 2–3 times smaller than in the regions close to the (20,20) CNT wall. This result is consistent with the stronger fluid interactions with the (20,20) CNT compared with the silica slit pores (Figure S2, Supporting Information). Finally, note that the density profiles for the IL in silica slit pores (Figure 1 and Figure S4, Supporting Information) are neither smooth nor completely symmetric relative to the center plane ($z = 0$). This indicates that longer simulations are required to get better statistics.

3.2.3. Interaction Energies. In addition to the density, the interactions between the IL molecules themselves and between the IL and the silica slit pores were calculated. The corresponding results are listed in Table S3 (Supporting Information). The bonded energy values, U_{IL}^b , for the bulk and confined IL are similar, suggesting that the intramolecular configurations for the bulk and confined IL are also similar. The nonbonded energy, $U_{\text{IL}}^{\text{nb}}$, for the confined $[\text{hmim}][\text{Tf}_2\text{N}]$ in the

(20,20) CNT is weaker than that in the silica slit pores, which is, in turn, weaker than $U_{\text{IL}}^{\text{nb}}$ for the bulk IL. This observation is consistent with the smaller density of the IL confined in the (20,20) CNT relative to that confined in the silica slit pores and the fact that the bulk IL exhibits the highest density (Figure S3, Supporting Information). The solid–fluid interaction energy, $U_{\text{sf}}^{\text{nb}}$, between the silica slit pore and the IL becomes weaker when the slit pore width is increased; it changes from -8.19 kJ/mol in the 25-Å-wide pore to -4.23 kJ/mol in the 45-Å-wide pore. The IL interaction with the (20,20) CNT is about 62 kJ/mol stronger than that with the 25-Å-wide silica slit pore (Table S3, Supporting Information), which is consistent with the stronger solid–fluid potential for the (20,20) CNT than for the silica pores (Figure S2, Supporting Information).

The total nonbonded energy, U_{t}^{nb} , for the bulk IL is 5–12 kJ/mol greater than that for the IL confined in silica slit pores and 30 kJ/mol smaller than that for the IL confined in the (20,20) CNT. These findings are consistent with the experimental observations that melting temperatures of ILs decrease when the ILs are confined in silica pores,^{33–35} but increase when the ILs are confined in a CNT.³⁶ It is expected that the melting temperature will decrease when the nonbonded interaction, U_{t}^{nb} , becomes weaker.

3.3. Thermodynamic Properties of CO₂, N₂, H₂, and H₂O Absorbed in Confined [hmim][Tf₂N]. **3.3.1. Solubilities of Gases in Confined [hmim][Tf₂N].** The calculated gas solubilities of CO₂, H₂, N₂, and H₂O in silica-confined [hmim][Tf₂N] are summarized in Table S2 (Supporting Information). The CO₂ solubilities in IL confined in a silica pore of 25-Å width are about 1.1–1.7 times larger than those in the bulk IL¹⁵ at 333–573 K and 4–83.6 bar (Figure S5, Supporting Information). In larger silica pores, the CO₂ solubilities in the confined IL are still typically larger than those in the bulk IL. For example, the CO₂ solubilities in the IL confined in the 35-Å-wide silica pore are 20–25% larger than those in the bulk IL at 333 K and 4–17.15 bar. In the 45-Å-wide silica slit pore, the CO₂ solubility in the confined IL is about 10% larger than that in the bulk IL at 333 K and 8 bar. As shown in Figure S5 (Supporting Information), it appears that CO₂ in the bulk IL generally shows a linear relationship between pressure and CO₂ solubility, but the relationship for the confined IL is noticeably nonlinear, especially at high pressures. When CO₂ molecules are absorbed in the IL, the cations and anions will adjust themselves to allow for more free volume for CO₂ molecules to fit. Under confinement, this adjustment is restricted by silica walls. Consequently, the CO₂ solubility in the confined IL is a nonlinear function of pressure and approaches a plateau-like region compared to that of the bulk IL. For H₂, the gas solubilities in the confined IL in silica pores of 25–45 Å are 1.2–2.3 times larger than those in the bulk IL²³ at 313–573 K and 50–300 bar. Similarly, the N₂ solubilities in the confined IL in a 25-Å-wide silica slit pore are 1.8–3.4 times larger than those in the bulk IL²² at 298–333 K and 40–500 bar. In contrast, the water solubilities in the confined and bulk ILs¹⁵ are comparable with an average difference of about 10% at 373 K in a 25-Å-silica slit pore.

When the silica slit pore width is increased, the CO₂ and H₂ solubilities in confined [hmim][Tf₂N] typically decrease (Table S2, Supporting Information). For example, when the silica slit pore width is increased from 25 to 35 Å, CO₂ solubilities decrease by about 20% at 333 K and 4–17 bar, and the H₂ solubilities decrease by about 30–45% at 313 K and 50–300 bar. When the silica slit pore width is further increased from 35

to 45 Å, CO₂ the solubilities decrease by 10–20% at 333 K and 4–17 bar, and the H₂ solubility decreases by about 15% at 313 K and 100 bar. In larger silica slit pores, confined [hmim][Tf₂N] exhibits smaller molar volumes (Figure S3, Supporting Information). This decrease results in lower solubilities for CO₂ and H₂ in IL confined in larger silica pores.

When the temperature is increased from 333 to 573 K, the CO₂ solubility in confined [hmim][Tf₂N] decreases (Table S2 and Figure S5, Supporting Information). In contrast, the H₂ solubility in the confined IL increases. Similarly to CO₂, the N₂ solubility in confined [hmim][Tf₂N] also decreases with temperature from 298 to 333 K. These temperature effects on the solubilities of CO₂, H₂, and N₂ in confined [hmim][Tf₂N] are the same as in the bulk IL.^{22,23} From the gas solubilities at different temperatures, the CO₂ heat of absorption in the 25-Å silica-pore-confined [hmim][Tf₂N] was estimated to be 10.6 ± 1.4 kJ/mol, very close to the value of 10.3 kJ/mol for CO₂ in the bulk IL.¹⁵

The interactions of CO₂, H₂, and N₂ with [hmim][Tf₂N] are weaker than the H₂O–[hmim][Tf₂N] interaction. For example, *ab initio* calculations give CO₂ interaction energies of about -17 and -22 kJ/mol with the [hmim]⁺ cation and [Tf₂N][−] anion, respectively. These interaction energies are much weaker than the corresponding H₂O interaction energies of -43 kJ/mol for both the [hmim]⁺ cation³⁷ and the [Tf₂N][−] anion through hydrogen bonding. For CO₂, H₂, and N₂, the IL molar volume is important in determining the gas solubility because of their weak interactions with [hmim][Tf₂N] compared to the water–[hmim][Tf₂N] interaction. Silica-confined [hmim][Tf₂N] exhibits larger molar volumes than the bulk IL (Figure S3, Supporting Information). Hence, the solubilities of CO₂, H₂, and N₂ in the silica-confined IL are 1.1–3.4 times larger than those in the bulk IL. As shown in Figure S3 (Supporting Information), [hmim][Tf₂N] confined in silica slit pores exhibits a much smaller molar volume than [hmim][Tf₂N] confined in a (20,20) CNT. Additionally, the fluid interaction with the (20,20) CNT is stronger than that with the silica pores (Figure S2, Supporting Information). Hence, the CO₂ and H₂ solubilities in 25-Å-silica-confined [hmim][Tf₂N] are 1.2–4 times smaller than those in the (20,20)-CNT-confined IL at 313–573 K and 4–200 bar.⁴ In the case of water, the molar volume of the IL plays a less important role in determining solubility. Hence, the gaseous water solubilities in silica-confined [hmim][Tf₂N] are only 10% different from the solubility values in the bulk IL.

The calculated CO₂ solubilities in silica-confined [hmim][Tf₂N] were compared with available experimental data for a similar system. Recently, Zhang et al.³⁸ experimentally determined that, for [bmim][BF₄] confined in silica gel pores of 30–120-Å width, the CO₂ solubilities in the confined [bmim][BF₄] were 1.5 times larger than that in the corresponding IL simply coated on the silica gel surface. Their experimental data suggest that the CO₂ solubility in the silica-confined IL is larger than that in the bulk IL, which is consistent with our simulations.

Finally, the degree of gas adsorption in empty silica slit pores was calculated (Table S1, Supporting Information) and compared with the gas solubilities in silica-confined [hmim][Tf₂N]. The H₂ solubilities in the confined IL are 4–13 times smaller than those in the empty silica pores (25–45 Å) at 313–573 K and 50–100 bar. The N₂ solubilities in the confined IL are 4–7 times smaller than those in the empty silica pore (25 Å) at 298–333 K and 50–500 bar. The larger amounts of H₂

and N_2 adsorbed in the empty silica pores compared to the confined IL are due to the larger void space (free volume) in the empty silica pores. For water, the confined IL exhibits gas solubilities about 40 times higher than those in the empty silica pore (25 Å) at 373 K and 0.05–0.2 bar. This increase is due to the much stronger interaction of water with $[\text{hmim}][\text{Tf}_2\text{N}]$ (Figure S6, Supporting Information) compared to the silica pore. The CO_2 solubility exhibits complicated behavior. At a low temperature of 333 K and pressures of 4–17.15 bar, the CO_2 solubilities in the confined IL are 1.4–2.0 times larger than those in the empty silica pores (25–45 Å). However, at high temperatures of 423–573 K and pressures of 8–83.6 bar, the CO_2 solubilities in the confined IL are 1.2–1.9 times smaller than those in the empty silica pore (25 Å).

3.3.2. Structures of Gases and Confined IL. Three representative snapshots of CO_2 , H_2 , and H_2O absorption in confined $[\text{hmim}][\text{Tf}_2\text{N}]$ are shown in Figure 2. Gas absorption in a confined IL is complex. One could envision a competition between absorption (in the IL volume) and adsorption (on the silica surface). At low gas concentrations, most H_2 molecules are positioned in the regions near the silica walls. In the case of CO_2 , some CO_2 molecules are absorbed in the regions near the wall, whereas other CO_2 molecules are absorbed inside the IL (Figures 2 and 3). Most N_2 molecules also occupy the regions near the wall at low gas solubilities (not shown). In contrast, water molecules are located inside the IL region rather than near the silica walls. This arrangement is due to the strong hydrogen-bonding interactions between water and $[\text{hmim}][\text{Tf}_2\text{N}]$ (Figure S6, Supporting Information). The corresponding local density distributions of the centers of mass for CO_2 , H_2 , N_2 , and H_2O absorption in $[\text{hmim}][\text{Tf}_2\text{N}]$ confined in a silica pore (25 Å) are shown in Figure 3.

3.4. Self-Diffusivity. **3.4.1. Self-Diffusivities of Gases in Confined $[\text{hmim}][\text{Tf}_2\text{N}]$ Using the Smooth Silica Potential Model.** The solid–fluid interaction energy in the smooth potential model depends only on z (eq 4), resulting in no net force in the xy plane for the adsorbate molecules. Therefore, the centers of mass of all the adsorbate molecules (IL + gas) were moved to the same origin in the xy plane when the mean squared displacement (MSD) was calculated. The MSD values in the xy plane for the centers of mass of CO_2 , H_2 , and N_2 gases at 8 mol % in the (25-Å) silica-pore-confined $[\text{hmim}][\text{Tf}_2\text{N}]$ are shown in Figure 4. The MSD values for CO_2 , H_2 , and N_2 in the confined IL are much larger than those in the corresponding bulk IL at 298–573 K. Gas self-diffusivity in the confined IL was calculated from the MSD value,²³ and the results are summarized in Table 2. The CO_2 self-diffusivity values in the confined IL are 2.4–8.3 times larger than those in the bulk IL at 313–573 K. The H_2 self-diffusivities in the confined IL are 3.7–4.5 times larger than those in the bulk IL at 313–573 K. The N_2 self-diffusivities in the confined IL are 3.4–5.3 times larger than those in the bulk IL at 298–373 K. Note that most $\beta = d(\log \Delta r^2)/d(\log t)$ values for all gases are close to 1 (Table 2), indicating that 10-ns NVE simulations are long enough to obtain gas normal diffusion.

The enhanced gas self-diffusivity values for CO_2 , H_2 , and N_2 gases in the confined $[\text{hmim}][\text{Tf}_2\text{N}]$ compared to the bulk IL are likely associated with these gases being mainly absorbed in the regions close to the silica wall where the IL density is very low (Figure 3a–c). It is expected that these gas molecules will experience far less frequent collisions with the IL molecules near the silica wall than in the bulk IL, which leads to larger self-diffusivity values for these gases in the silica-confined IL.

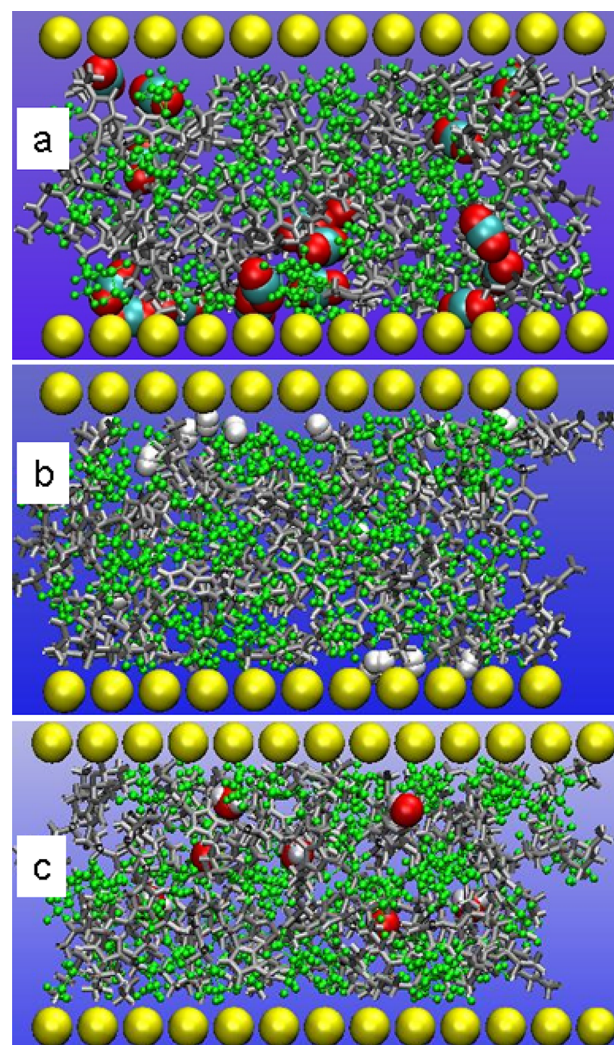


Figure 2. Snapshots of CO_2 , H_2 , and H_2O gaseous absorption in $[\text{hmim}][\text{Tf}_2\text{N}]$ confined in a 25-Å silica slit pore. Isostress-osmotic Monte Carlo simulations were implemented using the smooth silica potential model. The gas molecules are indicated using the van der Waals (vdW) graphical representation. The colors for gas molecules are as follows: red, oxygen atoms of CO_2 and H_2O ; cyan, carbon atoms of CO_2 ; white, hydrogen atoms of H_2 and H_2O . The $[\text{hmim}]^+$ cations (gray) and $[\text{Tf}_2\text{N}]^-$ anions (green) are shown using the bonds graphical representation. The silica walls are schematically represented by yellow spheres. (a) CO_2 absorption at 333 K and 8 bar; $L_x = L_y = 40.95$ Å. (b) H_2 absorption at 313 K and 100 bar; $L_x = L_y = 40.28$ Å. (c) H_2O absorption at 373 K and 0.2 bar; $L_x = L_y = 41.01$ Å.

Similarly, it has been shown that the self-diffusivity coefficients of CO_2 and H_2 gases in CNT-confined $[\text{hmim}][\text{Tf}_2\text{N}]$ are much larger than those in the bulk IL,⁴ partly because these gases mainly are positioned close to the CNT tube wall where the IL exhibits a very low density (not shown here).

In contrast, water in confined $[\text{hmim}][\text{Tf}_2\text{N}]$ exhibits diffusion behavior very different from that of CO_2 , H_2 , and N_2 gases. As shown in Figure S7 (Supporting Information), at 298 K, water in the confined IL exhibits MSD values much larger than those in the bulk IL, so that the self-diffusivity of water in the confined IL is 1.8 times larger than that in the bulk IL (Table 2). However, the MSD values for water in the confined and bulk ILs are comparable to each other at the higher temperature of 373 K. As discussed previously (Figure 3d), water exhibits the largest density inside the IL region

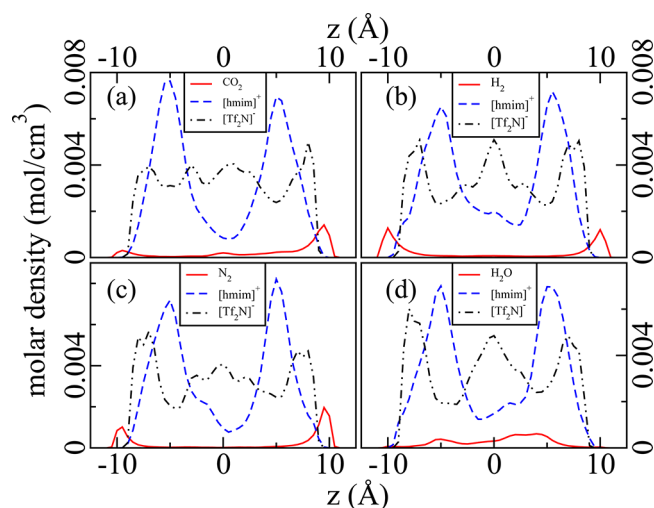


Figure 3. Molar density distributions of the centers of mass of CO₂, H₂, N₂, and H₂O gases; [hmim]⁺ cations; and [Tf₂N][−] anions confined in a 25-Å silica slit pore. The results were obtained from molecular dynamics simulations using the smooth silica potential model. (a) CO₂ absorption at 313 K, (b) H₂ absorption at 313 K, (c) N₂ absorption at 298 K, and (d) H₂O absorption at 298 K. Each system consists of five gas molecules and 60 pairs of cations and anions. The solid line in each panel indicates the density distribution of the gas; the dashed and dot-dashed lines are for the [hmim]⁺ cation and [Tf₂N][−] anion, respectively.

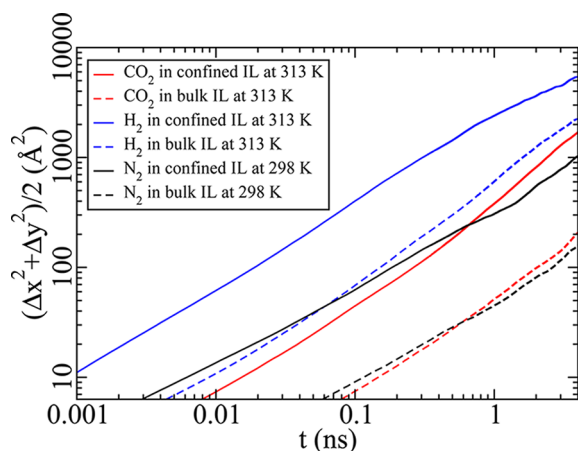


Figure 4. Mean squared displacement (MSD) in the *xy* plane of the centers of mass of CO₂, H₂, and N₂ gases in silica-pore-confined [hmim][Tf₂N] (solid lines). The MSD values were obtained from NVE molecular dynamics simulations using the smooth silica potential model for systems consisting of five gas and 60 ionic liquid (IL) molecules confined in a 25-Å silica slit pore. For comparison, MSD values for corresponding gas absorption in the bulk IL are also shown at low gas concentrations of 3 mol % (dashed lines). Legend: red solid and dashed lines, CO₂ at 313 K; blue solid and dashed lines, H₂ at 313 K; black solid and dashed lines, N₂ at 298 K.

where the IL density is also high. Hence, unlike CO₂, H₂, and N₂ gases, the enhanced water self-diffusivity in the confined IL at 298 K cannot be ascribed to the density distribution profiles of water and the confined IL. Water molecules form hydrogen bonds with both the [hmim]⁺ cations and the [Tf₂N][−] anions (Figure S6, Supporting Information), coupling the diffusivities of water and the IL. As shown in the next section, confined [hmim][Tf₂N] diffuses faster than the bulk IL at 298 K and, as

Table 2. Self-Diffusivity Coefficients (*D*) in the *xy* Plane for CO₂, H₂, N₂, and H₂O Gases Absorbed in [hmim][Tf₂N] Confined in a 25-Å Silica Slit Pore and in the Bulk Ionic Liquid (IL) at 3 mol %^{a-c}, along with Values of $\beta = d(\log \Delta r^2)/d(\log t)$ ^d

gas	<i>T</i> (K)	in confined IL		in bulk IL	
		<i>D</i> (m ² /s)	β	<i>D</i> (m ² /s)	β
CO ₂	313	2.16(5) × 10 ^{−9}	1.05(2)	2.59(4) × 10 ^{−10}	1.00(1)
CO ₂	373	3.06(2) × 10 ^{−9}	0.95(1)	1.26(1) × 10 ^{−9}	0.99(1)
CO ₂	573	2.78(6) × 10 ^{−8}	1.03(2)	8.4(4) × 10 ^{−9}	0.98(5)
H ₂	313	1.21(4) × 10 ^{−8}	0.79(2)	3.15(6) × 10 ^{−9}	0.98(2)
H ₂	373	3.07(2) × 10 ^{−8}	1.00(1)	8.3(2) × 10 ^{−9}	1.01(2)
H ₂	573	1.82(3) × 10 ^{−7}	1.03(2)	4.03(6) × 10 ^{−8}	1.00(1)
N ₂	298	1.39(5) × 10 ^{−9}	1.02(4)	2.6(1) × 10 ^{−10}	0.97(5)
N ₂	373	5.8(2) × 10 ^{−9}	0.97(3)	1.69(2) × 10 ^{−9}	1.02(1)
H ₂ O	298	1.49(5) × 10 ^{−10}	1.00(4)	8.5(2) × 10 ^{−11}	1.03(2)
H ₂ O	373	8.5(1) × 10 ^{−10}	0.94(1)	8.5(2) × 10 ^{−10}	1.00(2)

^aResults obtained from NVE molecular dynamics simulations using the smooth silica potential model. ^bSystem consisted of five gas molecules and 60 pairs of confined cations and anions. ^cAll self-diffusivities calculated in this work except for the self-diffusivity coefficients for CO₂ and H₂ in the bulk IL, which were obtained from a previous work.²³ ^dUncertainties in the last digit given in parentheses.

a result, increases the self-diffusivity of water in the confined IL compared to the bulk IL.

3.4.2. Self-Diffusivity of Confined [hmim][Tf₂N] Using the Smooth Silica Potential Model. The MSD values in the *xy* plane for the [hmim]⁺ cation and [Tf₂N][−] anion confined in the silica slit pore (25 Å) are shown in Figure S8 (Supporting Information). At a lower temperature of 298 K, the confined IL exhibits a larger MSD value (Figure S8a, Supporting Information) and a 1.7 times larger self-diffusivity than the bulk IL. In contrast, at a higher temperature of 373 K, the confined IL exhibits a smaller MSD value (Figure S8b, Supporting Information) and a 1.4–1.7 times smaller self-diffusivity than the bulk IL. At the intermediate temperature of 313 K, the MSD and self-diffusivity values of the confined IL are similar to those of the bulk IL. The enhanced self-diffusivity of the confined IL at 298 K is consistent with other simulations. Rajput et al. demonstrated that the IL molecules in the center of a graphite slit pore could diffuse faster than those in the bulk IL.⁶

A similar temperature-dependent confinement effect on IL self-diffusivity was also obtained experimentally by Iacob et al.⁷ At 182 K, [bmim][BF₄] confined in a silica nanopore (104 Å) was found to diffuse faster than the corresponding bulk IL. However, when the temperature was increased to about 250 K, the bulk IL diffused faster than the confined IL. Silica-pore-confined [hmim][Tf₂N] exhibits a larger molar volume and smaller density (Figure S3, Supporting Information) than the bulk IL, and in some regions, the local density of the confined IL is smaller than that of the bulk IL (Figure 1 and Figure S4, Supporting Information). The IL molecules in the low-density regions are expected to diffuse faster than those in the bulk IL. This local increase in diffusivity leads to an enhanced overall diffusivity of the confined IL at low temperatures. On the other hand, the silica wall retards IL movement, and this reduction in mobility might be responsible for the decrease in self-diffusivity of the confined IL at high temperatures. More studies are needed to fully understand the role of temperature in the self-diffusivity of confined ILs.

3.4.3. Self-Diffusivities of CO₂ and Confined [hmim][Tf₂N] Using the Atomistic Silica Potential Model. All of the preceding self-diffusivity values for the confined IL and gases were calculated using a smooth silica potential model (eq 4). It is important to determine whether the atomistic and smooth silica potential models give similar self-diffusivity results for the confined IL and gases.

In the atomistic silica potential model, the van der Waals (vdW) parameters and charges of the Si and O atoms from Cruz-Chu et al.³⁹ were tuned in this work to give CO₂ adsorption in the silica slit pore (25 Å) consistent with the values obtained from the smooth silica potential model. The vdW parameters and charges of Si and O atoms were chosen to be $\epsilon_{\text{Si}} = 0.0375$ kcal/mol, $\sigma_{\text{Si}} = 3.826$ Å, $q_{\text{Si}} = 0.1$, $\epsilon_{\text{O}} = 0.01875$ kcal/mol, $\sigma_{\text{O}} = 3.118$ Å, and $q_{\text{O}} = -0.05$. Note that the x , y , and z coordinates of the Si and O atoms of the silica slit pore (Figure S9, Supporting Information) in the atomistic model were fixed during simulations. These atomistic force-field parameters for the Si and O atoms give CO₂ adsorption that differs by 8–15% from that obtained using the smooth silica potential model at 333–423 K and 8–17.15 bar (Table S4, Supporting Information), a satisfactory level of agreement.

Local Density Distributions Using the Atomistic Silica Potential Model. Local density distributions of the gas and IL molecules were calculated because they are important in determining self-diffusivities. The density profiles are shown in Figure S10 (Supporting Information). Overall, the atomistic and smooth potential models were found to give similar local density distributions for [hmim]⁺ cations, [Tf₂N][−] anions, and CO₂ molecules. Similarly to the smooth silica potential model, the atomistic silica model also predicts that CO₂ molecules exhibit the highest peak density close to the silica walls where the IL density is low. This result is consistent with the simulated snapshot (Figure S9, Supporting Information).

Self-Diffusivities Using the Atomistic Silica Potential Model. The atomistic silica model gives CO₂ self-diffusivities (Table S5, Supporting Information) that differ by only 3–10% from those obtained with the smooth silica model. The CO₂ self-diffusivity coefficients from the atomistic silica model in the confined IL are 7.5 and 2.5 times larger than those in the bulk IL at 313 and 373 K, respectively. This difference is also partly due to CO₂ absorption close to the silica wall (Figure S10, Supporting Information). When self-diffusivities were computed, the centers of mass of the IL and CO₂ molecules were not moved to the same origin because the net forces on the adsorbate molecules in the xy plane due to the atomistic silica walls were not necessarily zero. Most $\beta = d(\log \Delta r^2)/d(\log t)$ values for CO₂ and the confined IL were found to be close to 1 (Table S5, Supporting Information), suggesting that 10-ns NVE simulations were long enough to obtain normal diffusion for both the gas and the IL.

The MSD values in the xy plane for confined [hmim][Tf₂N] in a 25-Å silica pore are shown in Figure S11 (Supporting Information), and the corresponding self-diffusivities are summarized in Table S5 (Supporting Information). At 313 K, the atomistic silica potential model gives much larger MSD values for the confined IL than the smooth silica potential model, even though the local density distributions for the two models are similar (Figure S10, Supporting Information). At a higher temperature of 373 K, the atomistic silica model gives slightly larger MSD values than the smooth silica potential model. These results suggest that the atomistic silica model has a more significant effect on the IL dynamics than on the IL

structure, especially at low temperatures. Despite the MSD difference between the two silica models, the atomistic silica model also predicts that the confined IL exhibits larger self-diffusivity at lower temperatures and smaller self-diffusivity at higher temperatures compared with the bulk IL. At a lower temperature of 313 K, the self-diffusivity of the confined IL obtained from the atomistic silica model is 4.7 times larger than that of the bulk IL. However, at 373 K, the self-diffusivity of the confined IL from the atomistic silica model is 1.3 times smaller than that of the bulk IL.

3.4.4. Effects of Silica Slit Pore Size on Self-Diffusivity. Self-diffusivities of CO₂ and [hmim][Tf₂N] confined in a larger silica slit pore (45 Å) were also calculated using the atomistic silica potential model and compared with the results obtained in a smaller silica pore (25 Å). As shown in Table S5 (Supporting Information), the CO₂ self-diffusivities in the 45-Å-pore-confined IL at 313–373 K are 30–43% smaller than those in the 25-Å-pore-confined IL, but still 1.8–4.3 times larger than the CO₂ diffusivity coefficients in the corresponding bulk IL. The behavior of the self-diffusivity coefficient of the confined IL in different pores as a function of temperature within the range of 313–373 K is complex. At 313 K, the self-diffusivity of the confined IL in the 45-Å pore is close to the value in the 25-Å pore and 5 times larger than the diffusivity of the bulk IL. However, at 373 K, the self-diffusivity of the confined IL in the 45-Å pore is 3.3 times larger than that in the 25-Å pore and 2.7 times larger than the diffusivity value of the bulk IL.

3.4.5. Effects of Silica Force-Field Parameters on Self-Diffusivity. Finally, the effects of the force-field parameters of Si and O atoms on self-diffusivity were examined. When the ϵ values for Si and O atoms in the atomistic silica model were increased by a factor of 8, leading to stronger solid–fluid interactions, the CO₂ self-diffusivity at 313 K in the confined IL (25 Å pore) decreased by 22% but remained 5.8 times larger than the CO₂ diffusivity in the bulk IL. Additionally, the CO₂ adsorption in an empty silica slit pore (25 Å) was found to increase by a factor of 6.4. In contrast, the self-diffusivity of the confined IL at 313 K decreased by a factor of 32, becoming 7.5 times smaller than the diffusivity of the bulk IL. These findings strongly suggest that the force-field parameters of the Si and O atoms affect the self-diffusivity of CO₂ in a small but significant way for the confined IL. It is interesting to note that all simulations in this work showed that the self-diffusivity of CO₂ in the confined IL is much larger than that in the bulk IL, regardless of whether the diffusivity of the confined IL is larger or smaller than that of the bulk IL.

4. CONCLUSIONS

Monte Carlo and molecular dynamics simulations were implemented to study pure [hmim][Tf₂N] confined in silica slit pores (25–45 Å). The absorption of gaseous CO₂, H₂, N₂, and H₂O in the confined IL was also examined, and self-diffusivity coefficients for both the gases and the IL molecules were determined.

Silica-confined [hmim][Tf₂N] exhibits a 12–31% smaller density than the corresponding bulk IL at 313–573 K and 1 bar. Several cation and anion layers occur in silica slit pores, and these layers exhibit peak densities about twice those in the bulk IL. The nonbonded interaction energy for bulk [hmim][Tf₂N] is about 5–12 kJ/mol larger than that for the silica-pore-confined IL but 30 kJ/mol smaller than that for the (20,20)-CNT-confined IL. These values are consistent with increased

melting temperatures for the CNT-confined IL³⁶ and decreased melting temperatures for the silica-pore-confined IL.^{33–35}

For gases that interact weakly with IL molecules, the molar volume of the confined IL is important in determining their solubilities. Larger molar volumes for the confined IL lead to solubilities of weakly absorbing gases such as CO₂, H₂, and N₂ in the silica-pore-confined [hmim][Tf₂N] that are 1.1–3 times larger than those in the bulk IL at 298–573 K. This increase is consistent with experimental findings showing that the CO₂ solubility in silica-pore-confined [bmim][BF₄] is larger than that in the corresponding bulk IL.³⁸ However, for water, which interacts strongly with [hmim][Tf₂N] through water–IL hydrogen-bonding interactions, the molar volume of the confined IL is less important in determining the water solubility. The water solubilities in silica-pore-confined and bulk [hmim][Tf₂N] were found to be similar, with a difference of less than 10%.

Simulations show that silica-pore-confined [hmim][Tf₂N] exhibits a larger self-diffusivity at 298 K but a smaller self-diffusivity at 373 K than the corresponding bulk IL. This result is consistent with the experimental findings of Iacob et al.⁷ that silica-pore-confined [bmim][BF₄] exhibits a much larger self-diffusivity at 182 K but a smaller self-diffusivity at 250 K than the bulk IL.

The self-diffusivities of weakly absorbing gases such as CO₂, H₂, and N₂ in (25-Å) silica-pore-confined [hmim][Tf₂N] were found to be 2.4–8.3 times larger than those in the bulk IL at 298–573 K. This trend arises because these gases are most likely absorbed close to the silica walls where the IL density is very low. Because of both the higher CO₂ solubility and the larger CO₂ self-diffusivity in the confined IL compared with the bulk, the CO₂ permeability in silica-pore-confined [hmim][Tf₂N] is expected to be larger than that in the bulk IL. These computational findings are consistent with the recent experimental work by Close et al. showing that the CO₂ permeability in an alumina-confined IL is larger than that in the bulk IL.⁸ In the case of water, because of strong water hydrogen-bonding interactions with [hmim][Tf₂N], water molecules are not absorbed close to the silica walls, and the water self-diffusivity is correlated with the IL dynamics. At 298 K, the water self-diffusivity in the confined IL was found to be 1.8 times larger than that in the bulk IL, partly because of the faster dynamics of the confined IL at 298 K. At 373 K, the water self-diffusivities in the confined and bulk ILs are comparable.

An atomistic silica potential model was also used to calculate the gas and IL self-diffusivities. The CO₂ self-diffusivity coefficients in the confined IL obtained from the atomistic and smooth silica potential models were similar. Although the atomistic silica potential model gives density distributions for the confined IL similar to those obtained using the smooth silica model, it gives a much larger self-diffusivity at 313 K and a slightly larger self-diffusivity at 373 K than the smooth silica model. Similarly to the smooth silica potential model, the atomistic silica model predicts a much larger CO₂ self-diffusivity in the confined IL than in the bulk IL; the confined IL exhibits larger self-diffusivities at lower temperatures but smaller self-diffusivities at higher temperatures than the bulk IL. Furthermore, when the solid–fluid interaction is increased by a factor of 8, the CO₂ self-diffusivity in the confined IL is still 5.8 times larger than that in the bulk IL, even though the confined IL itself exhibits a self-diffusivity that is 7.5 times smaller than that of the bulk IL.

■ ASSOCIATED CONTENT

■ Supporting Information

Details for the acceptance rule of area A change in $NP_{xy}T$ Monte Carlo simulations, to build the atomistic silica slit pores; validation of the FGSF method against the standard Ewald method to compute electrostatic interactions in two-dimensional silica slit pores; and adsorptions of CO₂, N₂, H₂, and H₂O gases in empty silica slit pores. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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